SOMMERS DUALITY AND THE SPRINGER–STEINBERG MAP FOR CANONICAL LEFT CELLS

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ABSTRACT. Let $(G, G^\vee)$ be the Langlands-dual pair of complex simple Lie groups and $\mathcal{O} \subset \mathrm{Lie}(G)$ be a special nilpotent orbit whose Barbasch–Vogan dual orbit $\mathcal{O}^\vee \subset \mathrm{Lie}(G^\vee)$ is even. We prove that Sommers' generalized duality map for $\mathcal{O}$ agrees with the Springer–Steinberg map on the diagonal $\Gamma \cap \Gamma^{-1}$ of the canonical Kazhdan–Lusztig left cell attached to $\mathcal{O}^\vee$.

1. INTRODUCTION

Let G and $G^\vee$ be Langlands dual connected complex reductive groups. Fix a maximal torus $T^\vee$ contained in a Borel subgroup $B^\vee$, and put

$$\mathfrak{g}^\vee = \mathrm{Lie}(G^\vee), \qquad W = N_{G^\vee}(T^\vee)/T^\vee.$$

Let $\mathfrak{n}^\vee$ be the nilradical of $\mathrm{Lie}(B^\vee)$. For $v \in W$, the Springer–Steinberg orbit $\mathrm{St}_{G^\vee}(v)$ is the unique nilpotent orbit meeting

$$\mathfrak{n}^\vee \cap \mathrm{Ad}(\dot{v})\mathfrak{n}^\vee$$

in a dense open subset, where $\dot{v}$ represents v in $N_{G^\vee}(T^\vee)$. Equivalently,

$$\overline{\mathrm{St}_{G^\vee}(v)} = \overline{G^\vee \cdot (\mathfrak{n}^\vee \cap \mathrm{Ad}(\dot{v})\mathfrak{n}^\vee)}. \tag{1}$$

In this convention, $\mathrm{St}_{G^\vee}(1)$ is the regular orbit.

Fix an even nilpotent orbit $\mathcal{O}^\vee$, and $(e^\vee, h^\vee, f^\vee)$ a Jacobson–Morozov triple for $\mathcal{O}^\vee$, where the semisimple element $h^\vee$ is dominant and the eigenvalues of $\mathrm{ad}(h^\vee)$ are all even. Let $L^\vee = Z_{G^\vee}(h^\vee)$, and let I be the set of simple reflections associated with the simple roots vanishing on $h^\vee$. Write $W_I = W(L^\vee)$ and let w_I be its longest element. Then the orbit $\mathcal{O}^\vee$ is the Richardson orbit of the corresponding parabolic subgroup. Consequently,

$$\mathrm{St}_{G^\vee}(w_I) = \mathcal{O}^\vee. \tag{2}$$

Write $\mathfrak{g} = \mathrm{Lie}(G)$ and put $\mathcal{O} = d_{BV}(\mathcal{O}^\vee) \subseteq \mathfrak{g}$, where d_{BV} denotes Barbasch–Vogan duality. Let $\overline{A}(\mathcal{O})$ be Lusztig's canonical quotient. Sommers' generalized duality [15, §6] associates to each conjugacy class $[x]$ in $\overline{A}(\mathcal{O})$ a nilpotent orbit

$$d_S(\mathcal{O}, [x]) = \mathcal{O}^\vee_{[x]} \subseteq \mathfrak{g}^\vee.$$

Definitions and conventions are recalled in Section 2.

Let Γ be the Kazhdan–Lusztig left cell containing w_I. We call this particular cell the **canonical left cell**. The correspondence in the following theorem is constructed by the inductions in Sections 6 and 7, with a choice of linked chain at each step.

Theorem 1.1. *With the notation above, there is a bijection*

$$\{[x] : x \in \overline{A}(\mathcal{O})\} \longrightarrow \Gamma \cap \Gamma^{-1}, \qquad [x] \longmapsto v_{[x]},$$

constructed as described below, such that

$$v_{[1]} = w_I, \qquad \mathrm{St}_{G^\vee}(v_{[x]}) = d_S(\mathcal{O}, [x]) \quad (x \in \overline{A}(\mathcal{O})). \tag{3}$$

When $\overline{A}(\mathcal{O})$ is trivial, the above theorem is given by (2) directly. In general, we first apply the results of Barbasch–Trapa [2] to prove the theorem when both $\mathcal{O}$ and $\mathcal{O}^\vee$ are even (Proposition 3.1). Afterwards, we prove that the equalities in (3) are preserved by smooth induction (Corollary 5.6). This induction step only requires the orbits in $\mathfrak{g}^\vee$ to be even, without assuming

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that the orbits in $\mathfrak{g}$ are even. This is used in Sections 6 and 7 to reduce every even $\mathcal{O}^\vee$ with nontrivial canonical quotient back to the Barbasch–Trapa setup.

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As for mathematical proofs, ChatGPT 6 Pro initially provided a proof only for Type B and C. The author then further asked for a possibility to follow the pathway of [3, Section 7] on the case when $\mathcal{O}^\vee$ is smoothly induced, and exploit the saturation-induction relations of nilpotent orbits for Sommers' duality in [14], which results in the content of Sections 4 and 5 below.

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2. Geometric and algebraic conventions

2.1. **Springer-Steinberg map.** Let G, $G^\vee$, $T^\vee$, $B^\vee$, and $\mathfrak{n}^\vee$ be as given in the introduction. We identify the Weyl groups of G and $G^\vee$ by corresponding root and coroot reflections. Write $\mathcal{B}^\vee = G^\vee/B^\vee$ and let $\mathcal{N}^\vee \subseteq \mathfrak{g}^\vee$ be the nilpotent cone. For $v \in W$, set

$$X_{G^\vee}(v) = \mathfrak{n}^\vee \cap \mathrm{Ad}(\dot{v})\mathfrak{n}^\vee.$$

For a standard Levi subgroup $M^\vee$ and $v \in W(M^\vee)$, define $X_{M^\vee}(v)$ in the same way using $B^\vee \cap M^\vee$. For a Borel subgroup B_i, let $\mathfrak{n}_{B_i}$ denote the nilradical of $\mathrm{Lie}(B_i)$. The Steinberg variety is

$$Z_{G^\vee} = \{(\xi, B_1, B_2) \in \mathcal{N}^\vee \times \mathcal{B}^\vee \times \mathcal{B}^\vee : \xi \in \mathfrak{n}_{B_1} \cap \mathfrak{n}_{B_2}\}.$$

Let Z_v be the closure of the locus where (B_1, B_2) has relative position v. The proper projection $Z_v \to \mathcal{N}^\vee$ has image $\overline{\mathrm{St}_{G^\vee}(v)}$ as given in [5, §2.1].

2.2. **Springer correspondence and nilpotent orbits.**

Definition 2.1 (Ordinary Springer support)**.** Let $\mathcal{P}^\vee \subseteq \mathfrak{g}^\vee$ be a nilpotent orbit with $e \in \mathcal{P}^\vee$, put $A_{G^\vee}(e) = Z_{G^\vee}(e)/Z_{G^\vee}(e)^\circ$. We write its Springer representation as

$$\mathrm{Spr}_{G^\vee}(\mathcal{P}^\vee, \rho)$$

where ρ is an irreducible representation of $A_{G^\vee}(e)$.

Conversely, for an irreducible W-representation R, there exists a unique $\mathcal{O}^\vee$ and $\rho \in \mathrm{Irr}\, A_{G^\vee}(e)$ such that $R \cong \mathrm{Spr}_{G^\vee}(\mathcal{O}^\vee, \rho)$. Then we write

$$\mathrm{Supp}_{G^\vee}(R) = \mathcal{O}^\vee.$$

We use the normalization

$$\mathrm{Spr}_{G^\vee}(\mathcal{O}^\vee_{\mathrm{reg}}, \mathbf{1}) = \mathbf{1}, \qquad \mathrm{Spr}_{G^\vee}(\{0\}, \mathbf{1}) = \mathrm{sgn}_W,$$

and the same convention for G and for Levi subgroups.

We say a nilpotent orbit $\mathcal{P}^\vee$ **special** if its corresponding Springer representation $\mathrm{Spr}_{G^\vee}(\mathcal{P}^\vee, \mathbf{1})$ is a special Weyl group representation in the sense of [9, Chapter 5]. For a special orbit $\mathcal{P}^\vee$, let $\mathscr{F}_{\mathcal{P}^\vee}$ be the Lusztig family containing its special Springer representation, and $\mathcal{S}_{\mathcal{P}^\vee}$ be the representations $\rho \in \mathrm{Irr}\, A_{G^\vee}(e)$ for which $\mathrm{Spr}_{G^\vee}(\mathcal{P}^\vee, \rho)$ occurs and belongs to $\mathscr{F}_{\mathcal{P}^\vee}$. Then

$$K_{G^\vee}(e) = \bigcap_{\rho \in \mathcal{S}_{\mathcal{P}^\vee}} \ker \rho, \qquad \overline{A}(\mathcal{P}^\vee) = A_{G^\vee}(e)/K_{G^\vee}(e).$$

This is Lusztig's canonical quotient [3, (4.4)].

Definition 2.2 (Barbasch–Vogan duality)**.** For a nilpotent orbit $\mathcal{P}^\vee \subseteq \mathfrak{g}^\vee$, let E be the unique special representation in the Lusztig family containing $\mathrm{Spr}_{G^\vee}(\mathcal{P}^\vee, \mathbf{1}) \otimes \mathrm{sgn}_W$. The **Barbasch–Vogan dual** of $\mathcal{P}^\vee$ is

$$d_{BV}(\mathcal{P}^\vee) = \mathrm{Supp}_G(E).$$

The same definition applies with G and $G^\vee$ interchanged.

Note that d_{BV} is an order-reversing involution between the *special* nilpotent orbits of $\mathfrak{g}$ and $\mathfrak{g}^\vee$. For *non-exceptional* orbits in the sense of [3, Definition 4.5] (which holds for all even orbits and their duals), tensoring with sgn_W takes the special Springer representation to that of the dual orbit, and their canonical quotients are isomorphic.

Definition 2.3. The **special piece** $\mathrm{SP}(\mathcal{O}^\vee)$ of a special orbit $\mathcal{O}^\vee$ consists of the nilpotent orbits whose least special orbit containing them in its closure is $\mathcal{O}^\vee$. Equivalently, it is $\overline{\mathcal{O}^\vee}$ with the closures of all strictly smaller special orbits removed [10, §0.1].

Definition 2.4 (Induced orbit and saturation)**.** Let $P^\vee = M^\vee U^\vee$ be a parabolic subgroup of $G^\vee$, with $\mathfrak{u}^\vee = \mathrm{Lie}(U^\vee)$, and let $\mathcal{E}$ be a nilpotent $M^\vee$-orbit. The **induced orbit** $\mathrm{Ind}_{M^\vee}^{G^\vee}(\mathcal{E})$ is the unique dense orbit in $G^\vee \cdot (\mathcal{E} + \mathfrak{u}^\vee)$. Its closure is the image of the proper map

$$G^\vee \times^{P^\vee} (\overline{\mathcal{E}} + \mathfrak{u}^\vee) \longrightarrow \mathfrak{g}^\vee.$$

For $e \in \mathcal{E}$, the orbit $G^\vee \cdot e$ is called the **saturation** of $\mathcal{E}$ in $G^\vee$.

We distinguish two operations on Weyl-group representations. The symbol J denotes truncated induction defined by Lusztig's a-invariant, as in [3, Definition 4.11]. The symbol j denotes truncated induction defined by the least degree of occurrence in the symmetric algebra of the reflection representation. For a representation occurring with multiplicity one in its least degree, the latter is the construction used in [15, §6].

Definition 2.5 (Smooth induction)**.** Let $M^\vee$ be a Levi subgroup of $G^\vee$, let $\mathcal{O}_M^\vee$ be special, and put $\mathcal{O}^\vee = \mathrm{Ind}_{M^\vee}^{G^\vee}(\mathcal{O}_M^\vee)$. Following [3, Definition 4.15], this induction is **smooth** if $J_{W(M^\vee)}^{W(G^\vee)}$ is a bijection between the families of $\mathcal{O}_M^\vee$ and $\mathcal{O}^\vee$. By [3, Proposition 4.16], this is equivalent to

$$\overline{A}(\mathcal{O}_M^\vee) \simeq \overline{A}(\mathcal{O}^\vee).$$

Definition 2.6 (Sommers duality)**.** Let $\mathcal{O}$ be a special nilpotent orbit of G, and let $[x]$ be a conjugacy class in $\overline{A}(\mathcal{O})$. Choose $e \in \mathcal{O}$ and a semisimple element $s \in Z_G(e)$ whose component has image in $[x]$. Put $L = Z_G(s)^\circ$. Write d_L for Lusztig–Spaltenstein duality within $\mathfrak{l} = \mathrm{Lie}(L)$, as in [15, §6]. Then **Sommers' dual map**

$$d_S(\mathcal{O}, [x]) \subseteq \mathfrak{g}^\vee$$

defines a nilpotent orbit in $\mathfrak{g}^\vee$, characterized by its Springer representation

$$\mathrm{Spr}_{G^\vee}(d_S(\mathcal{O}, [x]), \mathbf{1}) = j_{W(L)}^{W(G)} \mathrm{Spr}_L(d_L(L \cdot e), \mathbf{1}).$$

In particular, the map is independent of the choice of s and, in the special case when $[x] = [1]$ is the identity class, then $d_S(\mathcal{O}, [1]) = d_{BV}(\mathcal{O})$ is the usual Barbasch–Vogan dual.

Definition 2.7 (Linked Levi subgroups)**.** Two standard Levi subgroups are **adjacent** if their sets of simple roots agree outside a standard root subsystem of type A, their general linear blocks within that subsystem differ by an exchange of two adjacent blocks, and every common simple root outside the subsystem is orthogonal to it.

Two standard Levi subgroups are **linked** if they can be joined by a finite sequence of adjacent Levi subgroups; see [3, Definition 7.3]. A **linked chain** is a specified sequence of these exchanges.

3. Pairs of even dual orbits

The following result is a special case of Theorem 1.1, which follows from the results of Barbasch–Trapa [2]. We include the proof of it for the convenience of the reader:

Proposition 3.1. *Let G and $G^\vee$ be Langlands dual connected complex reductive groups. Let $\mathcal{O}^\vee \subseteq \mathfrak{g}^\vee$ be an even nilpotent orbit, and suppose that $\mathcal{O} = d_{BV}(\mathcal{O}^\vee)$ is also even. Let Γ be the left cell containing the longest element w_I of the Weyl group of the Dynkin Levi subgroup of $\mathcal{O}^\vee$. There is a bijection*

$$\{[x] : x \in \overline{A}(\mathcal{O})\} \longrightarrow \Gamma \cap \Gamma^{-1}, \qquad [x] \longmapsto v_{[x]},$$

such that

$$\mathrm{St}_{G^\vee}(v_{[x]}) = d_S(\mathcal{O}, [x]) \quad \text{for every } x \in \overline{A}(\mathcal{O}), \qquad v_{[1]} = w_I.$$

For each $[x]$, the element $v_{[x]}$ is the unique longest W_I-double-coset representative with the stated Springer–Steinberg orbit.

Proof. By [3, Theorem 4.7(d), 5.24–5.28] and [2, Proposition 3.4],

$$|\{[x] : x \in \overline{A}(\mathcal{O})\}| = |\operatorname{SP}(\mathcal{O}^\vee)| = |\Gamma \cap \Gamma^{-1}|.$$

Since $d_{BV}(\mathcal{P}^\vee) = \mathcal{O}$, every $\mathcal{P}^\vee \in \operatorname{SP}(\mathcal{O}^\vee)$ has a preimage $(\mathcal{O}, [x])$ under Sommers duality by [15, Propositions 15]. The first equality above implies that

$$[x] \mapsto d_S(\mathcal{O}, [x])$$

is a bijection onto $\operatorname{SP}(\mathcal{O}^\vee)$.

On the other hand, for each $\mathcal{P}^\vee \in \operatorname{SP}(\mathcal{O}^\vee)$, [2, Theorem 1.1(a)] gives a unique longest W_I-double-coset representative $v(\mathcal{P}^\vee)$ such that $\mathrm{St}_{G^\vee}(v(\mathcal{P}^\vee)) = \mathcal{P}^\vee$. We claim that

$$v(\mathcal{P}^\vee) \in \Gamma \cap \Gamma^{-1}.$$

Put $\lambda = h^\vee/2$, let w_0 be the longest element of W, and write $v = v(\mathcal{P}^\vee)$. We use the parameters $\overline{X}(\lambda, v\lambda)$ of [3, §5] and show that this representation is special unipotent. Indeed, let $P_I^\vee$ be the standard parabolic subgroup with Levi Weyl group W_I, and let $Q_{v(\mathcal{P}^\vee)}$ be the diagonal $G^\vee$-orbit on $G^\vee/P_I^\vee \times G^\vee/P_I^\vee$ corresponding to $W_I v(\mathcal{P}^\vee) W_I$. Let $\mathfrak{u}_I^\vee$ be the nilradical of $\mathrm{Lie}(P_I^\vee)$. Since $v(\mathcal{P}^\vee)$ is longest in its double coset, $v(\mathcal{P}^\vee)^{\pm 1}(\Phi_I^+) \subset \Phi^-$, where Φ_I^+ is the positive Levi root system. Hence

$$X_{G^\vee}(v(\mathcal{P}^\vee)) = \mathfrak{u}_I^\vee \cap \mathrm{Ad}(\dot{v})\mathfrak{u}_I^\vee.$$

The conormal moment image is the $G^\vee$-saturation of this space, so its dense orbit is $\mathcal{P}^\vee$.

Following [2, (4.4)], let $\pi(Q_{v(\mathcal{P}^\vee)})$ be the virtual character characterized by

$$\langle \pi(Q_{v(\mathcal{P}^\vee)}), [\mathcal{F}] \rangle_{\mathrm{ABV}} = m_{Q_{v(\mathcal{P}^\vee)}}(\mathcal{F}),$$

where $\langle\ ,\ \rangle_{\mathrm{ABV}}$ is the pairing between virtual characters and equivariant perverse sheaves, and $m_{Q_{v(\mathcal{P}^\vee)}}(\mathcal{F})$ is the coefficient of the conormal closure to $Q_{v(\mathcal{P}^\vee)}$ in the characteristic cycle of $\mathcal{F}$. As the conormal moment image has dense orbit $\mathcal{P}^\vee \in \operatorname{SP}(\mathcal{O}^\vee)$, [2, Proposition 4.3(2)] places $\pi(Q_v)$ in the integral span of special-unipotent representations attached to $\mathcal{O}^\vee$.

In these parameters, $\mathrm{IC}(\overline{Q_{v(\mathcal{P}^\vee)}}, \mathbf{1})$ is paired with $\overline{X}(\lambda, v(\mathcal{P}^\vee)\lambda)$. Thus $\overline{X}(\lambda, v\lambda)$ occurs in $\pi(Q_{v(\mathcal{P}^\vee)})$ with coefficient ± 1, and is special unipotent. Since $w_0\lambda = -\lambda$, w_0 normalizes W_I and commutes with w_I. Applying [3, Corollary 5.20 and (3.12)–(3.13)] to the regular lift with $w = w_0 v$ gives

$$w_0 v(\mathcal{P}^\vee) \sim_L w_0 w_I, \qquad w_0 v(\mathcal{P}^\vee) \sim_R w_0 w_I.$$

and hence

$$v(\mathcal{P}^\vee) \sim_L w_I, \qquad v(\mathcal{P}^\vee) \sim_R w_I.$$

Since $w_I = w_I^{-1}$, this proves the claim.

By the second equality in the beginning of the proof, the elements $v(\mathcal{P}^\vee)$ exhaust $\Gamma \cap \Gamma^{-1}$, and the assignment

$$v_{[x]} := v\big(d_S(\mathcal{O}, [x])\big).$$

gives the desired bijection. □

4. Smooth induction and Sommers duality

Let $M \subseteq G$ be a Levi subgroup and let $M^\vee \subseteq G^\vee$ be the corresponding dual Levi subgroup. Put $\mathfrak{m} = \mathrm{Lie}(M)$, $\mathfrak{m}^\vee = \mathrm{Lie}(M^\vee)$, $W_G = W(G)$, and $W_M = W(M)$. Fix special nilpotent orbits $\mathcal{O}_M^\vee \subseteq \mathfrak{m}^\vee$ and $\mathcal{O}^\vee \subseteq \mathfrak{g}^\vee$ such that

$$\mathcal{O}^\vee = \mathrm{Ind}_{M^\vee}^{G^\vee}(\mathcal{O}_M^\vee). \tag{4}$$

Assume that both orbits are even and that the induction is smooth, in the sense of Section 2. Set

$$\mathcal{O}_M = d_{BV}(\mathcal{O}_M^\vee), \qquad \mathcal{O} = d_{BV}(\mathcal{O}^\vee).$$

The orbits $\mathcal{O}_M$ and $\mathcal{O}$ need not be even.

We use the compatibility of Sommers duality with saturation and nilpotent induction in [14, Proposition 2.12(ii)]. By part (iii) of the same proposition, its specialization to the identity class is the corresponding Barbasch–Vogan identity.

Proposition 4.1. *For every $e \in \mathcal{O}_M$, one has $G \cdot e = \mathcal{O}$, and the inclusion $Z_M(e) \subseteq Z_G(e)$ induces an isomorphism*

$$\iota_e : \overline{A}(\mathcal{O}_M) \xrightarrow{\sim} \overline{A}(\mathcal{O}). \tag{5}$$

For every conjugacy class $[x_M]$ in $\overline{A}(\mathcal{O}_M)$,

$$d_S(\mathcal{O}, \iota_e([x_M])) = \mathrm{Ind}_{M^\vee}^{G^\vee} d_S(\mathcal{O}_M, [x_M]). \tag{6}$$

Proof. Put $\mathcal{O}' = G \cdot e$. As mentioned in the paragraph before the proposition, [14] gives

$$d_{BV}(\mathcal{O}') = \mathrm{Ind}_{M^\vee}^{G^\vee} d_{BV}(\mathcal{O}_M) = \mathcal{O}^\vee.$$

Set $E' = \mathrm{Spr}_G(\mathcal{O}', \mathbf{1})$. By Definition 2.2, $E' \otimes \mathrm{sgn}_{W_G}$ belongs to the family attached to $\mathcal{O}^\vee$.

Write $E_M = \mathrm{Spr}_M(\mathcal{O}_M, \mathbf{1})$. The geometric Springer restriction theorem [8, Theorem 8.3] gives

$$E_M \subseteq \mathrm{Res}_{W_M}^{W_G} E'.$$

Indeed, choose a parabolic subgroup $P = MU$ and put $\mathfrak{u} = \mathrm{Lie}(U)$. The variety $(G \cdot e) \cap (e + \mathfrak{u})$ contains $U \cdot e$, and

$$d := \dim(U \cdot e) = \frac{\dim(G \cdot e) - \dim(M \cdot e)}{2},$$

because $\mathfrak{u}$ and its opposite are dual modules for an $\mathfrak{sl}_2$-triple of e in $\mathfrak{m}$, so $\dim((G \cdot e) \cap (e + \mathfrak{u})) \geq d$. Combined with [8, Proposition 1.2(a)], one has

$$\dim((G \cdot e) \cap (e + \mathfrak{u})) = d$$

Put

$$Y = \{g \in G : \mathrm{Ad}(g^{-1})e \in e + \mathfrak{u}\}.$$

The map $g \mapsto \mathrm{Ad}(g^{-1})e$ has fibers isomorphic to $Z_G(e)$, so Y has components of dimension $\dim Z_G(e) + d$. The action $(z, z_M) \cdot g = z g z_M^{-1}$ induces an $A_G(e) \times A_M(e)$-action on these components. Their sum is a nonzero jointly invariant vector in the corresponding permutation representation. The restriction formula (see also [13, 0.4 (4)]), applied to $\mathbf{1} \boxtimes \mathbf{1}$, therefore gives the asserted constituent with trivial local systems on both orbits.

By Barbasch–Vogan duality, $E_M \otimes \mathrm{sgn}_{W_M}$ is the special representation attached to $\mathcal{O}_M^\vee$, and $E' \otimes \mathrm{sgn}_{W_G}$ belongs to the family attached to $\mathcal{O}^\vee$. Frobenius reciprocity implies that the latter occurs in $\mathrm{Ind}_{W_M}^{W_G}(E_M \otimes \mathrm{sgn}_{W_M})$. Since the induction is smooth, [3, Propositions 4.14 and 4.16] say that the only constituent of this induced representation in that family is $\mathrm{Spr}_{G^\vee}(\mathcal{O}^\vee, \mathbf{1})$, i.e.

$$E' \otimes \mathrm{sgn}_{W_G} = \mathrm{Spr}_{G^\vee}(\mathcal{O}^\vee, \mathbf{1}), \qquad E' = \mathrm{Spr}_G(\mathcal{O}, \mathbf{1}).$$

Injectivity of the Springer correspondence gives $\mathcal{O}' = \mathcal{O}$.

For (5), the centralizer inclusion induces an injective map ι_e by [14, (2.1.2), Proposition 2.15 and Lemma 2.16(i),(iii)], which leaves the canonical quotients and their natural maps unchanged by [14, §2.10]. Smoothness gives

$$|\overline{A}(\mathcal{O}_M)| = |\overline{A}(\mathcal{O}_M^\vee)| = |\overline{A}(\mathcal{O}^\vee)| = |\overline{A}(\mathcal{O})|,$$

and ι_e is an isomorphism. Finally, Equation (6) follows directly from [14, Proposition 2.12(ii) and Corollary 2.14]. □

5. Induction of the Springer–Steinberg orbit

Let $M \subset G$ be an ordinary Levi subgroup, and let $M^\vee \subset G^\vee$ be the corresponding standard Levi subgroup. Suppose that

$$\mathcal{O}^\vee = \mathrm{Ind}_{M^\vee}^{G^\vee}(\mathcal{O}_M^\vee) \tag{7}$$

is smooth and that both $\mathcal{O}_M^\vee$ and $\mathcal{O}^\vee$ are even. Let I_M and I be their Dynkin zero sets. Let $\Gamma_M \subset W(M^\vee)$ be the left cell containing w_{I_M}, and let $\Gamma_* \subset W(G^\vee)$ be the left cell containing the same element.

Lemma 5.1. *Under these hypotheses,*

$$\Gamma_* \cap \Gamma_*^{-1} = \Gamma_M \cap \Gamma_M^{-1}$$

as subsets of $W(G^\vee)$.

Proof. Parabolic Kazhdan–Lusztig multiplication gives $\Gamma_M \subset \Gamma_*$ and hence $\Gamma_* \cap \Gamma_*^{-1} \supseteq \Gamma_M \cap \Gamma_M^{-1}$. By [3, Proposition 5.28], the canonical cell representation $[\Gamma_M]$ is a multiplicity-free sum of r distinct irreducible representations. Apply [3, Proposition 4.14(b)] to $w = w_{I_M}$ in $W(M^\vee)$. It identifies the cell modules containing this same element in the two Weyl groups:

$$[\Gamma_*] = J_{W(M^\vee)}^{W(G^\vee)}[\Gamma_M].$$

Smoothness and [3, Proposition 4.16] therefore imply that $[\Gamma_*]$ is also a sum of r distinct irreducible representations, each with multiplicity one. For every left cell Λ, [3, Proposition 5.25] gives $|\Lambda \cap \Lambda^{-1}| = \dim \mathrm{End}_W[\Lambda]$. Both sets in the inclusion therefore have cardinality r. □

Lemma 5.2. *Let* $P^\vee = M^\vee U^\vee \subset G^\vee$ *be standard and let* $u \in W(M^\vee)$. *Then*

$$X_{G^\vee}(u) = \mathrm{Lie}(U^\vee) \oplus X_{M^\vee}(u), \tag{8}$$

and

$$\mathrm{St}_{G^\vee}(u) = \mathrm{Ind}_{M^\vee}^{G^\vee} \mathrm{St}_{M^\vee}(u). \tag{9}$$

Proof. Every element of $W(M^\vee)$ preserves the positive roots of $G^\vee$ outside the Levi root system. Indeed, reflection in a simple Levi root does not alter the coefficients of the other simple roots. A positive root outside the Levi subsystem has a positive coefficient at one of those other roots and therefore cannot become negative. The root-space decomposition proves (8).

Let $\mathcal{E} = \mathrm{St}_{M^\vee}(u)$. By definition,

$$\overline{M^\vee \cdot X_{M^\vee}(u)} = \overline{\mathcal{E}}.$$

Since $M^\vee$ normalizes $\mathrm{Lie}(U^\vee)$, taking $G^\vee$-saturations and closures in (8) gives

$$\overline{G^\vee \cdot X_{G^\vee}(u)} = \overline{G^\vee \cdot (\mathcal{E} + \mathrm{Lie}(U^\vee))}.$$

The unique dense orbits in these two closed sets give (9). □

Lemmas 5.1 and 5.2 identify the inherited diagonal $\Gamma_M \cap \Gamma_M^{-1}$ with $\Gamma_* \cap \Gamma_*^{-1}$ and compute its Springer–Steinberg orbits by induction. The required correspondence takes values in the diagonal of the canonical left cell containing w_I. We therefore seek a bijection between these diagonals that sends w_{I_M} to w_I and preserves the Springer–Steinberg orbit in $G^\vee$. Under the linked-Levi hypothesis, Proposition 5.5 constructs such a bijection by a sequence of Kazhdan–Lusztig star operations. We first recall their definition from [6, §4].

Definition 5.3 (Star operations of type A_2)**.** Let α, β be simple roots such that the reflections $s = s_\alpha$ and $t = s_\beta$ generate a Weyl group of type A_2, equivalently, st has order three. Put

$$D_R(s,t) = \{v \in W : \text{exactly one of } vs < v \text{ and } vt < v \text{ holds}\}.$$

For $v \in D_R(s,t)$, exactly one of vs, vt belongs to $D_R(s,t)$. The **right star operation of type** A_2 is the involution $v \mapsto v^*$ defined by

$$\{v^*\} = \{vs, vt\} \cap D_R(s,t).$$

Equivalently, in each coset $d\langle s,t\rangle$ with d of minimal length, this operation exchanges

$$ds \longleftrightarrow dst, \qquad dt \longleftrightarrow dts.$$

The **left star operation** is defined on $D_R(s,t)^{-1}$ by ${}^*v = ((v^{-1})^*)^{-1}$.

Lemma 5.4. *A left or right star operation of type A_2 preserves the Springer–Steinberg orbit.*

Proof. Use the notation of Definition 5.3, and first consider the pair ds, dst. Let $\mathfrak{u}^\vee_{\alpha,\beta}$ be the nilradical of the standard parabolic subgroup whose Levi root system is generated by α and β. Put

$$V_d = \mathfrak{n}^\vee \cap \mathrm{Ad}(\dot{d})\mathfrak{u}^\vee_{\alpha,\beta}.$$

Minimality of d implies that $d\alpha, d\beta, d(\alpha+\beta)$ are positive and hence

$$X_{G^\vee}(d) = V_d \oplus \mathfrak{g}^\vee_{d\alpha} \oplus \mathfrak{g}^\vee_{d\beta} \oplus \mathfrak{g}^\vee_{d(\alpha+\beta)}.$$

A direct root-space calculation gives

$$X_{G^\vee}(ds) = V_d \oplus \mathfrak{g}^\vee_{d\beta} \oplus \mathfrak{g}^\vee_{d(\alpha+\beta)}, \qquad X_{G^\vee}(dst) = V_d \oplus \mathfrak{g}^\vee_{d\beta}.$$

The root subgroup $U^\vee_{d\alpha}$ preserves V_d: it preserves $\mathfrak{n}^\vee$ and, being in the conjugate Levi subgroup, normalizes $\mathrm{Ad}(\dot{d})\mathfrak{u}^\vee_{\alpha,\beta}$. Choose root vectors with

$$[e_{d\alpha}, e_{d\beta}] = c e_{d(\alpha+\beta)}, \qquad c \neq 0.$$

For $z \in V_d$ and $a \neq 0$, conjugation by $\exp(-b e_{d\alpha}/(ca)) \in U^\vee_{d\alpha}$ sends

$$z + a e_{d\beta} + b e_{d(\alpha+\beta)}$$

into $X_{G^\vee}(dst)$ since $2\alpha+\beta$ is not a root. Thus a dense open subset of $X_{G^\vee}(ds)$ is conjugate into $X_{G^\vee}(dst)$. The reverse inclusion of saturated closures follows from $X_{G^\vee}(dst) \subset X_{G^\vee}(ds)$. Hence one obtains

$$\mathrm{St}_{G^\vee}(ds) = \mathrm{St}_{G^\vee}(dst).$$

Interchanging α and β proves the other right string. Finally,

$$\mathrm{Ad}(\dot{v}^{-1})X_{G^\vee}(v) = X_{G^\vee}(v^{-1})$$

reduces left star operations to right star operations. □

Proposition 5.5. *Assume that the induction in* (7) *is smooth, $\mathcal{O}^\vee_M$ and $\mathcal{O}^\vee$ are even, and that their Dynkin Levi subgroups are linked. Let Γ be the left cell containing w_I. Then there is a finite sequence of left and right star operations of type A_2 which sends w_{I_M} to w_I.*

The same sequence is defined on every element of $\Gamma_M \cap \Gamma_M^{-1}$, and its composition gives a bijection

$$F : \Gamma_M \cap \Gamma_M^{-1} \longrightarrow \Gamma \cap \Gamma^{-1}$$

such that $F(w_{I_M}) = w_I$ and

$$\mathrm{St}_{G^\vee}(F(u)) = \mathrm{Ind}_{M^\vee}^{G^\vee} \mathrm{St}_{M^\vee}(u) \quad (u \in \Gamma_M \cap \Gamma_M^{-1}). \tag{10}$$

If $I_M = I$, then F is the inclusion in Lemma 5.1.

Proof. Fix a linked chain from I_M to I, and write the longest Weyl group elements of their corresponding Levi subgroup as $w_{I_M}, \dots, w, w', \dots, w_I$.

Consider the adjacent longest elements w and w'. Take J to span just the two exchanged blocks. By the definition of adjacency, we may write

$$w = ca, \qquad w' = cb,$$

where a, b are the longest elements of Young subgroups in a standard type-A subgroup W_J, and c is the longest element of the common factor orthogonal to J.

Under the Robinson–Schensted algorithm, the shapes of a and b coincide. If the exchanged blocks have sizes p, q, then $W_J = S_{p+q}$, and a, b are the longest elements of $S_p \times S_q$ and $S_q \times S_p$, respectively. Both shapes have column lengths p, q, arranged in decreasing order.

Write $\mathrm{RS}(a) = (P(a), Q(a))$, $\mathrm{RS}(b) = (P(b), Q(b))$, and let $z_0 \in W_J$ with

$$\mathrm{RS}(z_0) = (P(b), Q(a)).$$

Knuth's theorem [7, Theorem 6] gives a sequence of left star operations from a to z_0, since their recording tableaux coincide, and a sequence of right star operations from z_0 to b, since their insertion tableaux coincide. On three consecutive entries, Because c commutes with W_J and $\ell(cx) = \ell(c) + \ell(x)$ for every $x \in W_J$, these operations remain defined after multiplication by c. Thus, putting $z = cz_0$, we obtain

$$w \xrightarrow{\text{left stars}} z \xrightarrow{\text{right stars}} w'.$$

Let Λ and Λ' be the left cells containing w and w', respectively, and denote the two compositions by F_L and F_R. We show that they induce bijections

$$\Lambda \cap \Lambda^{-1} \xrightarrow{F_L} \Lambda \cap (\Lambda')^{-1} \xrightarrow{F_R} \Lambda' \cap (\Lambda')^{-1}.$$

By [6, §4.1 and Corollary 4.3], a left star operation preserves the left cell of each element. Moreover, its domain is determined by the left descent set, which is constant on right cells, and it maps a right cell bijectively onto the right cell of its image. The corresponding statements for right star operations follow by inversion.

Since w, w' are involutions and the right star sequence preserves right cells, the right cells of w and z are Λ^{-1} and $(\Lambda')^{-1}$, respectively. The preceding properties therefore show that F_L maps these right cells bijectively while preserving each left cell. Restricting to Λ gives the first displayed bijection. And the bijection F_R follows analogously.

Compose these bijections along the linked chain. This gives

$$F : \Gamma_* \cap \Gamma_*^{-1} \xrightarrow{\sim} \Gamma \cap \Gamma^{-1}, \qquad F(w_{I_M}) = w_I.$$

Lemma 5.1 identifies the domain with $\Gamma_M \cap \Gamma_M^{-1}$. If $I_M = I$, the empty chain gives the asserted inclusion.

Every operation in the construction preserves the Springer–Steinberg orbit by Lemma 5.4. Hence Lemma 5.2 gives, for every $u \in \Gamma_M \cap \Gamma_M^{-1}$,

$$\mathrm{St}_{G^\vee}(F(u)) = \mathrm{St}_{G^\vee}(u) = \mathrm{Ind}_{M^\vee}^{G^\vee} \mathrm{St}_{M^\vee}(u),$$

as required. □

Corollary 5.6. *Under the hypotheses of Proposition 5.5, let $[x_M] \mapsto u_{[x_M]}$ be a bijection satisfying Theorem 1.1 for $\mathcal{O}_M^\vee$. Then*

$$v_{\iota_e([x_M])} = F(u_{[x_M]})$$

defines a bijection satisfying Theorem 1.1 for $\mathcal{O}^\vee$, where $e \in d_{BV}(\mathcal{O}_M^\vee)$ and ι_e is the isomorphism of Proposition 4.1.

Proof. Propositions 4.1 and 5.5 give the bijectivity and

$$\mathrm{St}_{G^\vee}(F(u_{[x_M]})) = \mathrm{Ind}_{M^\vee}^{G^\vee} \mathrm{St}_{M^\vee}(u_{[x_M]}) = d_S(\mathcal{O}, \iota_e([x_M])).$$

The identity class is preserved because $\iota_e([1]) = [1]$ and $F(w_{I_M}) = w_I$. □

6. Even nilpotent orbits in the classical groups

In this section, we study the case when $G^\vee$ is classical. For an even orbit $\mathcal{O}^\vee$ with nontrivial $\overline{A}(\mathcal{O}^\vee)$, we construct a pair of even dual orbits $(\mathcal{O}_\delta, \mathcal{O}_\delta^\vee)$ and obtain $\mathcal{O}^\vee$ by successive smooth inductions on $\mathcal{O}_\delta^\vee$.

Let $X \in \{B, C, D\}$ be the type of $G^\vee$. Throughout this section, we identify $\mathcal{O}^\vee$ with its partition using column sizes as in [17, Section 2.4]. For a partition μ of type X, write $\mathcal{O}_\mu^\vee$ for the corresponding orbit and $|\mu|$ for its number of boxes.

An orbit $\mathcal{O}^\vee$ is even if and only if all its positive rows have the same parity. If the partition for $\mathcal{O}^\vee$ is all odd in type C, or all even in type D (the *very-even* orbits), $\overline{A}(\mathcal{O}^\vee)$ is trivial. For the remaining even orbits, the row sizes are odd in types B, D and even in type C. Thus the columns occur in equal pairs, apart from one column of maximal height m in types B, D, where m is odd in type B and even in type D.

For $\mathcal{O}^\vee = \mathcal{O}_\mu^\vee$, let $\mathcal{I}(\mu)$ be the set of distinct positive column heights that are odd in type B and even in types C, D, excluding the maximal column height in types B, D. The column description in [17, Proposition 2.4] gives

$$\overline{A}(\mathcal{O}^\vee) \simeq (\mathbb{Z}/2\mathbb{Z})^{|\mathcal{I}(\mu)|}. \tag{11}$$

Assume $\mathcal{I}(\mu) \neq \emptyset$. Let δ be the partition given by

$$\delta := \begin{cases} (m) \cup \bigcup_{a \in \mathcal{I}(\mu)} (a, a), & X = B, D, \\ \bigcup_{a \in \mathcal{I}(\mu)} (a, a), & X = C. \end{cases} \tag{12}$$

After removing the columns δ from μ, the remaining columns occur in pairs. So μ can be expressed in the form

$$\mu = \delta \cup \bigcup_{i=1}^{t} (k_i, k_i). \tag{13}$$

for integers $k_1 \geq k_2 \geq \cdots \geq k_t > 0$.

For example, let $G^\vee$ be of type C, and take $\mathcal{O}^\vee = \mathcal{O}_\mu^\vee$ with $\mu = (6, 6, 4, 4, 4, 4, 3, 3, 1, 1)$. The distinct even column heights are 4 and 6, so

$$\mathcal{I}(\mu) = \{4, 6\}, \qquad \delta = (6, 6, 4, 4), \qquad (k_1, k_2, k_3) = (4, 3, 1).$$

Lemma 6.1. *The partition δ has type X, and both $\mathcal{O}_\delta^\vee$ and $d_{BV}(\mathcal{O}_\delta^\vee)$ are even. Moreover, $\mathcal{I}(\delta) = \mathcal{I}(\mu)$.*

Proof. This follows directly from the characterization of classical nilpotent orbits and their Lusztig's canonical quotients in [17, Section 2]. □

Proposition 6.2. *Let $\mathcal{O}^\vee$ be an even nilpotent orbit of a group of type B, C, or D, and suppose that $\overline{A}(\mathcal{O}^\vee)$ is nontrivial. Then $\mathcal{O}^\vee$ is obtained by successive smooth inductions from an even orbit $\mathcal{O}_\delta^\vee$ whose dual $\mathcal{O}_\delta$ is also even, satisfying the linked-Levi hypothesis of Corollary 5.6.*

Proof. Consider δ and the column pairs (k_i, k_i) as above. At every intermediate partition

$$\mu_r := \delta \cup (k_1, k_1) \cup \cdots \cup (k_r, k_r)$$

with $0 \leq r \leq t$, the column heights determining $\mathcal{I}$ are precisely those already present in δ. Thus $\mathcal{I}(\mu_r) = \mathcal{I}(\mu)$, and (11) gives the same canonical quotient at every step.

Let $G_{\mu_r}^\vee$ and $G_{\mu_r,k}^\vee$ have type X and defining dimensions $|\mu_r|$ and $|\mu_r| + 2k$, respectively. The classical induction formula [4, Corollary 7.3.4(ii)] gives

$$\mathcal{O}_{\mu_r \cup (k,k)}^\vee = \mathrm{Ind}_{GL_k \times G_{\mu_r}^\vee}^{G_{\mu_r,k}^\vee} (\{0\} \times \mathcal{O}_{\mu_r}^\vee). \tag{14}$$

The column pair (k, k) preserves the required row parity, with zero rows adjoined before adding the pair when necessary, so the partition on the left already has type X. The zero orbit of GL_k has trivial canonical quotient; hence each induction is smooth by [3, Proposition 4.16].

For these partitions, each equal column pair of height a contributes a factor GL_a to the Dynkin Levi subgroup. In types B, D, the unpaired column of height m contributes SO_m; in type C, there is no remaining classical factor. Indeed, these factors are the centralizers of the semisimple element of the $\mathfrak{sl}_2$-triple on its pairs of opposite nonzero eigenspaces and on its zero eigenspace, respectively. Adding (k, k) therefore adds exactly the factor GL_k. The inherited Dynkin Levi subgroup and the new Dynkin Levi subgroup differ only in the order of their general linear factors, so adjacent exchanges give a linked chain. Lemma 6.1 supplies the initial pair of even dual orbits. □

Corollary 6.3. *Theorem 1.1 holds in all classical Lie types.*

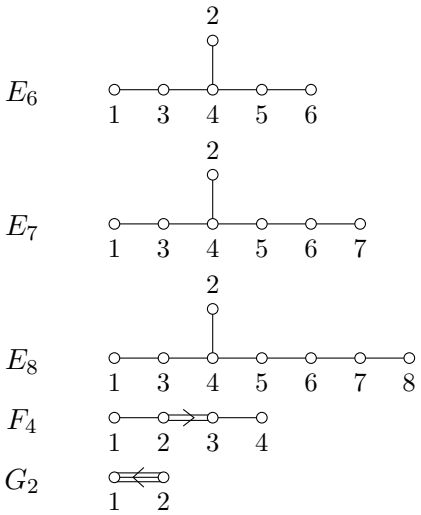
Proof. If $\overline{A}(\mathcal{O}^\vee)$ is trivial, $\Gamma \cap \Gamma^{-1} = \{w_I\}$ by [3, Propositions 5.25 and 5.28]. Then the result follows directly from (2). This includes all even type-A orbits, the all-odd partitions in type C, and the very-even partitions in type D.

When $\overline{A}(\mathcal{O}^\vee)$ is nontrivial, $\mathcal{O}^\vee$ has the form considered above. Apply Proposition 3.1 to $\mathcal{O}_\delta^\vee$ in Lemma 6.1, and then apply Corollary 5.6 at every induction of Proposition 6.2.

At each step, apply Corollary 5.6 to (w_{GL_k}, u_μ), where w_{GL_k} is the longest element of $W(GL_k)$ and u_μ is supplied by the correspondence already constructed for $\mathcal{O}_\mu^\vee$. □

7. The exceptional groups

Let $G^\vee$ be a simple complex group of exceptional type, and let G be its Langlands dual. We use the following Bourbaki numbering for the simple roots of $G^\vee$.



For G, we use the corresponding dual simple roots with the same indices; thus the arrows in types F_4 and G_2 are reversed.

We use the standard Bala–Carter symbols of [4, Chapter 8] and [1] for complex nilpotent orbits for exceptional groups. As discussed in the previous sections, it suffices to study the cases when $\mathcal{O}^\vee$ is even while $\mathcal{O}$ is not, and $\overline{A}(\mathcal{O}^\vee)$ is nontrivial. The weighted Dynkin diagrams and dual orbits in [4, Chapter 8] and the five exceptional tables of [1], together with the canonical quotients in [15, §9], give exactly nine such orbits, listed in Table 1 below.

In each row of Table 1, K specifies the simple roots of the chosen standard Levi subgroup $M^\vee$, and

$$\mathcal{O}^\vee = \mathrm{Ind}_{M^\vee}^{G^\vee}(\mathcal{O}_M^\vee).$$

Both $\mathcal{O}_M^\vee$ and $d_{BV}(\mathcal{O}_M^\vee)$ are even, and the Dynkin Levi subgroups of $\mathcal{O}_M^\vee$ and $\mathcal{O}^\vee$ coincide. Their common zero set $I = I_M$ is displayed in the table. These induction identities use the Richardson property of even orbits and induction by stages [4, Section 7.1]. The canonical quotients of the smaller and larger orbits are isomorphic to the group in the last column, so the inductions are smooth by [3, Proposition 4.16].

Corollary 7.1. *Theorem 1.1 holds for every even exceptional orbit, with the correspondence obtained from Proposition 3.1 and the inductions of Table 1.*

TABLE 1. Even $\mathcal{O}^\vee$ with nontrivial $\overline{A}(\mathcal{O}^\vee)$ and noneven $\mathcal{O}$, together with their inducing data. Here $\mathcal{P}_r^\vee$ denotes the orbit with row partition $[3^2, 1^{2r-6}]$ in $\mathfrak{so}_{2r}$.

	$G^\vee$	$\mathcal{O}^\vee$	$\mathcal{O}$	K	$I = I_M$	$\mathcal{O}_M^\vee$	$\overline{A}(\mathcal{O}^\vee)$
1	F_4	$F_4(a_1)$	$\widetilde{A}_1$	$\{2,3\}$	$\{3\}$	$B_2 : [3,1,1]$	S_2
2	E_7	$E_6(a_3)$	$D_4(a_1) + A_1$	$\{1,2,3,4,5,7\}$	$\{1,2,4,5,7\}$	$\mathcal{P}_5^\vee \times \{0\}$	S_2
3	E_7	A_4	$D_5(a_1)$	$\{2,3,4,5,6,7\}$	$\{2,3,4,5,7\}$	$\mathcal{P}_6^\vee$	S_2
4	E_7	$E_6(a_1)$	$A_2 + A_1$	$\{2,3,4,5,7\}$	$\{2,3,5,7\}$	$\mathcal{P}_4^\vee \times \{0\}$	S_2
5	E_8	$E_8(a_6)$	$D_4(a_1) + A_1$	$\{1,2,3,4,5,6,8\}$	$\{1,2,3,5,6,8\}$	$D_4(a_1) \times \{0\}$	S_3
6	E_8	A_4	$E_7(a_3)$	$\{1,2,3,4,5,6,7\}$	$\{2,3,4,5,6,7\}$	A_2	S_2
7	E_8	$E_6(a_3)$	$D_6(a_1)$	$\{2,3,4,5,6,7,8\}$	$\{2,3,4,5,6,8\}$	$\mathcal{P}_7^\vee$	S_2
8	E_8	$E_6(a_1)$	$D_5(a_1)$	$\{2,3,4,5,6,7\}$	$\{2,3,4,5,7\}$	$\mathcal{P}_6^\vee$	S_2
9	E_8	$E_8(a_4)$	$A_2 + A_1$	$\{2,3,4,5,7\}$	$\{2,3,5,7\}$	$\mathcal{P}_4^\vee \times \{0\}$	S_2

Proof. If $\overline{A}(\mathcal{O}^\vee)$ is trivial, the canonical diagonal consists of the longest element of the Dynkin Levi Weyl group, by [3, Propositions 5.25 and 5.28], and its Springer–Steinberg orbit is the Richardson orbit $\mathcal{O}^\vee$. Proposition 3.1 treats each pair of even dual orbits. The inductions in Table 1 satisfy the hypotheses of Corollary 5.6. Since the two sets of zero nodes coincide, conormal induction uses the inclusion of the smaller diagonal into the ambient Weyl group. Applying Corollary 5.6 once in each row proves the assertion. □

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